

# UMEA, Sweden

WMO: 022860

Lat: 63.792N

Lon: 20.283E

Elev: 46

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -10.8      | -5.5 | -16.4                         | 2.3 | -10.2 | -10.7 | 3.1 | -5.1 | 25.5                  | 31.5 | 23.0 | 31.3 | 2.9                   | 330  | 0.675 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.2                   | 76.5            | 63.5 | 73.3 | 61.6 | 70.1 | 60.3 | 66.0                | 72.7 | 63.9 | 70.4 | 61.9 | 67.8 | 8.6                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 63.2                            | 86.8 | 69.3 | 61.1 | 80.5 | 66.9 | 59.2 | 75.2 | 64.5 | 30.8          | 72.6 | 29.2 | 70.8 | 27.7 | 67.7 | 72.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 21.1          | 18.6 | 16.3 | -17.1                      | 81.0 | 6.9                | 4.4 | -22.0                                              | 84.1 | -26.1      | 86.7 | -30.0      | 89.2 | -35.0      | 92.4 |    |    |
| (5)               |      |      | -17.5                      | 68.1 | 6.8                | 2.5 | -22.4                                              | 69.9 | -26.4      | 71.4 | -30.2      | 72.8 | -35.2      | 74.7 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 38.2   | 19.5  | 18.9  | 25.6  | 35.2  | 44.9  | 54.2  | 60.1  | 57.4  | 49.2  | 38.6  | 30.0  | 23.7  |  |  |
|                                                                        | DBStd       | 16.34  | 11.36 | 12.09 | 9.10  | 5.88  | 6.49  | 5.19  | 4.75  | 5.22  | 5.75  | 6.83  | 8.46  | 10.61 |  |  |
|                                                                        | HDD50       | 5081   | 946   | 872   | 755   | 444   | 186   | 19    | 0     | 5     | 83    | 354   | 600   | 817   |  |  |
|                                                                        | HDD65       | 9798   | 1411  | 1292  | 1220  | 894   | 624   | 326   | 165   | 242   | 474   | 818   | 1050  | 1282  |  |  |
|                                                                        | CDD50       | 780    | 0     | 0     | 0     | 0     | 27    | 144   | 314   | 234   | 59    | 2     | 0     | 0     |  |  |
|                                                                        | CDD65       | 20     | 0     | 0     | 0     | 0     | 0     | 1     | 13    | 6     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 215    | 0     | 0     | 0     | 0     | 4     | 25    | 144   | 42    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 23     | 0     | 0     | 0     | 0     | 0     | 2     | 20    | 1     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 7.0    | 7.2   | 7.2   | 7.4   | 6.9   | 7.5   | 7.4   | 6.7   | 6.3   | 6.6   | 7.0   | 7.1   | 7.0   |  |  |
|                                                                        | PrecAvg     | 26.1   | 2.2   | 1.5   | 1.5   | 1.3   | 1.6   | 2.0   | 3.0   | 3.3   | 2.4   | 2.5   | 2.6   | 2.2   |  |  |
| (15) Precipitation                                                     | PrecMax     | 40.6   | 4.5   | 3.3   | 3.0   | 3.8   | 2.8   | 3.9   | 7.3   | 6.1   | 5.1   | 5.2   | 5.9   | 5.2   |  |  |
|                                                                        | PrecMin     | 17.6   | 0.9   | 0.1   | 0.4   | 0.3   | 0.2   | 0.7   | 0.3   | 0.7   | 0.6   | 0.9   | 0.7   | 0.4   |  |  |
|                                                                        | PrecStd     | 5.0    | 0.9   | 0.9   | 0.7   | 0.9   | 0.7   | 0.7   | 1.8   | 1.6   | 1.3   | 1.1   | 1.5   | 1.1   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 42.4  | 43.9  | 48.3  | 60.2  | 73.0  | 77.2  | 82.8  | 78.4  | 67.8  | 56.1  | 48.8  | 43.2  |  |  |
|                                                                        |             | MCWB   | 38.4  | 38.5  | 40.5  | 48.0  | 56.9  | 60.1  | 65.9  | 65.5  | 59.0  | 50.2  | 45.4  | 40.4  |  |  |
|                                                                        | 2%          | DB     | 38.3  | 40.0  | 44.3  | 53.7  | 66.9  | 72.0  | 77.7  | 74.6  | 64.4  | 53.3  | 45.7  | 40.8  |  |  |
|                                                                        |             | MCWB   | 35.8  | 35.9  | 37.7  | 43.9  | 54.3  | 58.5  | 64.5  | 63.3  | 56.6  | 49.5  | 44.1  | 38.9  |  |  |
|                                                                        | 5%          | DB     | 36.0  | 36.7  | 41.0  | 49.8  | 62.0  | 68.5  | 74.7  | 71.4  | 62.2  | 51.3  | 43.1  | 39.0  |  |  |
|                                                                        |             | MCWB   | 34.3  | 33.8  | 35.6  | 41.8  | 50.9  | 56.3  | 63.0  | 62.2  | 56.0  | 48.7  | 41.7  | 37.5  |  |  |
|                                                                        | 10%         | DB     | 34.0  | 34.1  | 37.5  | 46.1  | 57.9  | 65.9  | 71.5  | 68.1  | 59.5  | 49.4  | 41.3  | 37.2  |  |  |
|                                                                        |             | MCWB   | 32.8  | 32.7  | 33.4  | 39.3  | 47.8  | 54.9  | 61.4  | 60.2  | 54.6  | 47.0  | 40.0  | 35.9  |  |  |
|                                                                        | 0.4%        | WB     | 38.5  | 39.0  | 41.2  | 48.7  | 58.9  | 63.2  | 69.3  | 68.7  | 61.1  | 53.0  | 46.2  | 40.9  |  |  |
|                                                                        |             | MCDB   | 40.8  | 43.3  | 47.4  | 59.0  | 70.3  | 72.4  | 77.2  | 74.4  | 64.7  | 54.5  | 47.4  | 41.9  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 36.2  | 36.5  | 38.3  | 44.9  | 55.1  | 60.3  | 66.8  | 65.7  | 58.9  | 51.0  | 44.2  | 39.1  |  |  |
|                                                                        |             | MCDB   | 37.7  | 39.3  | 43.3  | 51.7  | 65.3  | 68.3  | 74.0  | 71.7  | 62.5  | 52.3  | 45.3  | 40.1  |  |  |
|                                                                        | 5%          | WB     | 34.4  | 34.2  | 36.0  | 42.2  | 51.9  | 58.3  | 65.0  | 63.4  | 57.2  | 49.0  | 42.1  | 37.3  |  |  |
|                                                                        |             | MCDB   | 35.7  | 35.5  | 39.7  | 48.6  | 60.4  | 65.8  | 71.9  | 68.8  | 60.6  | 50.5  | 43.0  | 38.3  |  |  |
|                                                                        | 10%         | WB     | 32.8  | 32.7  | 34.2  | 39.8  | 49.3  | 56.6  | 63.0  | 61.4  | 55.6  | 47.0  | 40.1  | 35.7  |  |  |
|                                                                        |             | MCDB   | 33.8  | 33.8  | 36.7  | 45.0  | 56.2  | 63.7  | 69.7  | 66.2  | 58.8  | 48.8  | 41.1  | 36.9  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 13.3  | 15.2  | 16.1  | 16.0  | 18.2  | 17.9  | 17.2  | 17.4  | 16.5  | 12.8  | 10.1  | 12.1  |  |  |
|                                                                        |             | MCDBR  | 12.4  | 15.3  | 17.0  | 23.5  | 25.6  | 23.5  | 23.6  | 21.7  | 19.5  | 15.5  | 10.2  | 10.5  |  |  |
|                                                                        |             | MCWBR  | 11.3  | 12.7  | 12.1  | 15.0  | 14.4  | 12.2  | 12.6  | 12.6  | 13.3  | 11.9  | 9.3   | 9.7   |  |  |
|                                                                        | 5% WB       | MCDBR  | 11.8  | 12.2  | 15.3  | 21.7  | 24.1  | 20.0  | 19.9  | 18.7  | 16.3  | 12.5  | 9.3   | 10.2  |  |  |
|                                                                        |             | MCWBR  | 11.0  | 10.6  | 11.2  | 14.5  | 14.4  | 11.7  | 11.8  | 12.0  | 11.9  | 10.7  | 8.9   | 9.7   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.224 | 0.286 | 0.311 | 0.348 | 0.350 | 0.349 | 0.367 | 0.355 | 0.326 | 0.310 | 0.360 | 0.350 |  |  |
|                                                                        | taud        |        | 2.025 | 2.175 | 2.251 | 2.319 | 2.450 | 2.489 | 2.462 | 2.490 | 2.551 | 2.403 | 2.648 | 2.608 |  |  |
|                                                                        | Ebn at Noon |        | 115   | 199   | 246   | 263   | 274   | 276   | 267   | 260   | 246   | 195   | 108   | 59    |  |  |
|                                                                        | Edh at Noon |        | 9     | 19    | 27    | 32    | 30    | 30    | 30    | 27    | 21    | 16    | 8     | 5     |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 42    | 225   | 701   | 1211  | 1612  | 1818  | 1664  | 1260  | 741   | 285   | 68    | 19    |  |  |
|                                                                        | RadStd      |        | 4     | 43    | 78    | 113   | 159   | 141   | 156   | 113   | 79    | 53    | 13    | 2     |  |  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (42 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page